

GUSTAVO SCETTINI

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EDUCATION

Virginia Polytechnic Institute and State University, Virginia Tech

Ph.D. in progress in Animal Sciences

2022 – Present

Blacksburg-VA, USA

Sao Paulo State University, UNESP

M.Sc. in Genetics and Animal Breeding

2019 – 2021

Jaboticabal-SP, Brazil

Federal University of Bahia, UFBA

B.S. in Veterinary Medicine

2013 – 2018

Salvador-BA, Brazil

COMPLEMENTARY EDUCATION

Graduate Teaching Assistant Workshop, Virginia Polytechnic Institute and State University.

2022

Systems Biology and Gene Networks Inference: Application to Livestock Breeding and Genetics, Sao Paulo University (ESALQ-USP).

2019

Data Analysis in R Software, Sao Paulo State University (UNESP).

2019

Genomic Selection Applied in Livestock: Methods and Tools, Sao Paulo State University (UNESP).

2019

Introduction to Using Software R, Federal University of Bahia (UFBA).

2015

TEACHING EXPERIENCE

Teaching Assistant. Course: Genomics (APSC 4054). Virginia Polytechnic Institute and State University.

Instructor: Fernando Biase

2023-2024

Teaching Assistant. Course: Animal Breeding and Genetics (ALS 3104). Virginia Polytechnic Institute and State University.

Instructor: Gota Morota

2023

Teaching Assistant. Course: Special Topics in Biotechnology in Animal Production (MEVB10). Federal University of

Bahia (UFBA). Instructor: Barbara M. P. da Silva Souza

2020

RESEARCH EXPERIENCE

Graduate Research Assistant, Virginia Polytechnic Institute and State University

2022- Present

Graduate Research Assistant, Sao Paulo State University (UNESP)

2019-2021

Undergraduate Research Internship, Federal University of Bahia (UFBA)

2015-2018

AWARDS

1. AG2PI Student Conference Travel Award (\$1500)

2023

RESEARCH SUPPORT

1. Graduate Research Development Program (\$1000)

09/13/2024–01/31/2025

Bovine *in vitro*-produced embryo morphokinetics profile associated with embryo competence.

PI: Schettini, G.P.

2. AG2PI support to develop genomic workflows (\$8000)

09/30/2023–05/31/2024

Integrating Illumina and ONT sequence workflow.

PI: Schettini, G.P.

3. Graduate Research Development Program (\$1000)

09/30/2022–10/01/2023

Transcriptome profile of *in vitro*-produced bovine blastocyst associated with hatching rate.

PI: Schettini, G.P.

PEER-REVIEWED JOURNAL PAPERS

1. Schettini G. P.; Morozyuk, M.; Biase, F.H. Identification of novel cattle (*Bos taurus*) genes and biological insights of their

function in pre-implantation embryo development. BMC Genomics, v. 25, 775, 2024. [DOI](#)

2. Kalbfleisch, T.S., McKay, S.D., Murdoch, B.M. *et al.* **The Ruminant Telomere-to-Telomere (RT2T) Consortium.** Nature Genetics, v. 56, 1566–1573, 2024. [DOI](#)
3. Biase F. H.; Schettini G. P. **Protocol for the electroporation of CRISPR-Cas for DNA and RNA targeting in Bos taurus zygotes.** STAR Protocols, v. 5, 2024. [DOI](#)
4. Nix, J. L.; Schettini, G. P.; Speckhart S. L.; Ealy, A. D.; Biase, F. H. **Ablation of OCT4 function in cattle embryos by double electroporation of CRISPR-Cas for DNA and RNA targeting (CRISPR-DART).** PNAS Nexus, pgad343, 2023. [DOI](#)
5. Nix, J. L.; Schettini, G. P.; Biase, F. H. **Sexing of cattle embryos using RNA-sequencing data or polymerase chain reaction based on a complete sequence of cattle chromosome Y.** Frontiers in Genetics, v. 14, 2023. [DOI](#)
6. Toro-Ospina, A.; Herrera Rios, A. C.; dos Santos, W. B.; Schettini, G. P.; Vallejo Aristizabal, V. H.; Tovar Claros, G.; Ortiz Morea, E. G. **Genetic Architecture and Signatures of Selection in the Caqueteño Creole (Colombian Native Cattle).** Diversity, v. 14, 828, 2022. [DOI](#)
7. Toro-Ospina, A.; Herrera Rios, A. C.; Schettini, G. P.; Vallejo Aristizabal, V. H.; dos Santos, W. B.; Zapata, C. A.; Ortiz Morea, E. G. **Identification of Runs of Homozygosity Islands and Genomic Estimated Inbreeding Values in Caqueteño Creole Cattle (Colombia).** Genes, v. 7, 1232, 2022. [DOI](#)
8. Schettini, G. P.; Peripolli, E.; Alexandre, P. A.; Dos Santos, W. B.; Pereira, A. S. C.; De Albuquerque, L. G.; Baldi, F.; Curi, R. A. **Transcriptome Profile Reveals Genetic and Metabolic Mechanisms Related to Essential Fatty Acid Content of Intramuscular Longissimus tho- racis in Nellore Cattle.** Metabolites, v. 12, 471, 2022. [DOI](#)
9. Schettini, G. P.; Peripolli, E.; Alexandre, P. A.; Dos Santos, W. B.; da Silva Neto, J. B.; Pereira, A. S. C.; De Albuquerque, L. G.; Curi, R. A.; Baldi, F. **Transcriptomic profile of longissimus thoracis associated with fatty acid content in Nellore beef cattle.** Animal Genetics, v. 53, p. 264-280, 2022. [DOI](#)
10. Dos Santos, W. B.; Schettini, G. P.; Maiorano, A. M.; Bussiman, F. O.; Balieiro, J. C. C.; Ferraz, G. C.; Pereira, G. L.; Baldassini, W. A.; Neto, O. R. M.; Neto, H. N. O.; Curi, R. A. **Genome-wide scans for signatures of selection in Mangalarga Marchador horses using high-throughput SNP genotyping.** BMC genomics, v. 22, p. 737-754, 2021. [DOI](#)
11. Schettini, G. P.; Lambert, S. M.; Da Silva Souza, B. M. P.; Costa, R. B.; De Camargo, G. M. F. **Genetic potential of Sindhi cattle for A2 milk production.** Animal Production Science, v. 60, p. 893-895, 2020. [DOI](#)
12. Dos Santos, W. B.; Schettini, G. P.; Fonseca, M. G.; Pereira, G. L.; Chardulo, L. A. L.; Neto, O.; Baldassini, W. A.; Oliveira, H. N.; Curi, R. A. **Fine-scale estimation of inbreeding rates, runs of homozygosity and genome-wide heterozygosity levels in the Mangalarga Marchador horse breed.** Journal of Animal Breeding and Genetics, v. 00, p. jbg.12508, 2020. [DOI](#)
13. Da Silva Neto, J. B.; Peripolli, E.; Da Costa e Silva, E. V.; Espigolan, R.; Neira, J. D. R.; Schettini, G. P.; Da Costa Filho, L. C. C.; Barbosa, F. B.; Macedo, G. G.; Costa-Brunes, L.; Lobo, R. B.; Pereira, A. S. C.; Baldi, F. **Genetic correlation estimates between age at puberty and growth, reproductive, and carcass traits in young Nelore bulls.** Livestock Science, v. 241, p.104266, 2020. [DOI](#)

SKILLS

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| <ol style="list-style-type: none">1. Laboratory-related techniques<ul style="list-style-type: none">• DNA and RNA extraction• Library construction for Next Generation Sequencing• Oxford Nanopore Technologies Sequencing• <i>In vitro</i> embryo production (bovine) | <ol style="list-style-type: none">2. Programming<ul style="list-style-type: none">• Unix/Linux• Bash• Awk• R• Markdown• Docker/Conda• Git |
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LANGUAGES

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| <ol style="list-style-type: none">1. English – Full professional proficiency2. Spanish – Professional working proficiency | <ol style="list-style-type: none">3. Portuguese – Native |
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REFERENCES

1. Prof. Dr. Fernando Biase – Associate Professor at Virginia Tech – fbias@vt.edu
2. Prof. Dr. Fernando Baldi Rey – Associate Professor at Sao Paulo State University – fernandobaldiuy@gmail.com